
Knowing Before Saying: Internal Correctness Signals in Mathematical Reasoning

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Abstract

1 When a language model is confidently wrong, is the error undetected or left
2 unsaid? We look inside the model to find out. Across four open models
3 and four domains (MMLU-Pro, MATH, GPQA-Diamond, and a geometry
4 task graded by a compiler), a simple linear detector reading the model’s
5 internal state predicts whether an answer is correct better than the model’s
6 own stated confidence in 12 of 16 settings, by a mean of +0.09 AUROC.
7 The gap is widest on MATH (+0.20). The detector is not just spotting
8 hard questions: given the same question several times, it still separates
9 the attempts that succeeded from the ones that failed. On Mistral, the
10 signal exists before we ask for it: on MMLU-Pro, a probe fitted after the
11 attempt reads at chance before it, and at the last token of the answer, with
12 no confidence question in play, correctness is readable at 0.69 against 0.57
13 from the question alone. And on Mistral, though not on GLM, the model’s
14 report depends on it: remove that direction after the answer is written and
15 stated confidence can no longer tell the model’s successes from its failures
16 ($0.84 \rightarrow 0.56$), while removing a random direction of the same size changes
17 nothing; turn it up and confidence on wrong answers drops while calibration
18 error halves. Models carry more about their own errors than they report,
19 and on at least one model, what they report depends on it.

20 1 Introduction

21 A language model that is confidently wrong is a problem. Its stated confidence is a signal
22 for deciding whether to trust its answer. This paper asks whether the model’s activations
23 contain the signal to know when it might be wrong.

24 Two things could be going on. Upon giving an incorrect answer and saying it has high
25 confidence, either the model genuinely has no idea it might be wrong and is just blindly
26 confident, or it deep down holds the error signal in its activations but unsurfaced to its
27 confidence report. Hence it is either a capability problem or a reporting problem.

28 In this paper we aim to tell these two apart. We read the model’s internal state directly
29 with a linear detector, a *probe*, and compare what the probe finds to the model’s reported
30 confidence. We say a fact is *represented* when a probe can read it from the model’s state,
31 and *reported* when it reaches the confidence number the model writes. The title’s “knowing”
32 is shorthand for a representation that is present, specific to the attempt, and, in at least one
33 model, used by the report. We make no claim about awareness, and a probe alone could not
34 support one.

35 Prior work has shown that a model’s output probabilities are often better calibrated than its
36 words [4], and that the truth of a given statement can be read from activations [1, 2, 5]. We
37 apply those tools to the model’s own attempts, and then intervene on the representation [6]
38 to ask whether the report depends on it.

39 2 Reading the state

40 **Models and tasks.** We tested four open models built in four different ways: Mistral-
41 Small-24B (dense), Qwen3.6-27B (gated linear attention interleaved with full attention), and
42 GLM-4.7-Flash and Gemma-4-26B (mixture of experts). Each answered 150 questions, five
43 times each, in four domains: MMLU-Pro, MATH, GPQA-Diamond, and GeoGenBench, a
44 geometry task in which the model writes a construction and a compiler checks it against a
45 formal specification, so its labels need no judge; in a pre-registered check of 200 attempts by
46 two trained raters, the compiler never failed a correct construction. That is 16 cells of about
47 750 attempts each.

48 Prompting structure.

- 49 1. The model is shown the question and asked, before answering, how confident it is, from 0
50 to 100, that it will get it right.
- 51 2. It answers.
- 52 3. It is asked how confident it is that the answer it just gave is correct.

53 **The probe.** At the moment the model is about to write its confidence digit, we record
54 the number and the residual-stream hidden state at that token (2,048 to 5,120 dimensions,
55 depending on the model). The probe is a linear classifier on that vector: features are
56 standardized, reduced to 50 principal components, and fed to an L2-regularized logistic
57 regression that predicts whether the attempt was correct. We fit it with five-fold cross-
58 validation grouped by question, so all five attempts at a question fall in the same fold and
59 the probe is always scored on questions it never saw. The read depth is fixed in advance at
60 70% of the network (layer 28 of 40 for Mistral, 33 of 47 for GLM, 21 of 30 for Gemma, 45 of
61 64 for Qwen3.6). Every readout is scored with AUROC.

62 **The activation state carries more than self-report.** The probe beats the model’s
63 stated confidence in **12 of 16** cells, with one tie. The mean gain is +0.09 AUROC (Ap-
64 pendix D). The gap is largest on MATH, at +0.20. Mistral on MATH is a good example
65 of what this looks like. The model is never told whether it got the answer right. Yet after
66 attempting, it raises its confidence, and it raises it more on the attempts it got wrong (+22
67 points) than on the ones it got right (+16 points). The probe on the same records reads
68 0.83. The state tracks correctness better than the self-report does.

69 **Why MATH.** On MATH, right and wrong answers have the same shape. Both are a
70 chain of steps ending in a boxed result, so the model cannot tell from the look of its answer
71 whether it went wrong. Appendix A follows one attempt where a single wrong subtraction
72 leads to a wrong answer, and the model then raises its confidence. Shape still carries a little
73 signal. A baseline classifier that sees only surface features of the answer text, such as its
74 length and line count, reaches 0.74, but the probe reaches 0.83. Geometry is the opposite
75 case. A failed construction usually does not compile, so the failure is visible in the answer
76 itself, and there the surface baseline does as well as the probe. The internal signal matters
77 most where the failure is invisible from outside.

78 **Controls.** The token we read is identical in every record, so at the input layer the probe
79 sits at exactly 0.5 and anything it finds deeper was computed during the attempt. A
80 within-question comparison holds difficulty fixed. For time, the controls below were run on
81 Mistral only. Its four cells reproduce the headline, with a smaller margin over the surface
82 baseline (+0.21 on MMLU-Pro, +0.07 on MATH, +0.05 on GPQA, where neither reads far
83 above chance, and −0.11 on geometry).

Table 1: Mistral. A probe trained on the row’s domain, tested on the column’s. Geo = GeoGenBench.

train ↓ test →	Geo	MMLU-Pro	MATH	GPQA
Geo	0.68	0.67	0.78	0.56
MMLU-Pro	0.70	0.75	0.84	0.59
MATH	0.80	0.72	0.83	0.57
GPQA	0.60	0.69	0.74	0.57

3 What the signal is

Difficulty control. A probe that beats stated confidence could be doing something simpler. It could be learning which questions are hard. Hard questions fail more often, so a difficulty detector would predict correctness without reading the attempt. We run two checks.

First, hold the question fixed. Each question is attempted five times, so within one question, difficulty cannot vary. The probe still separates the successful attempts from the failed ones (0.72 to 0.74 in representative cells). The model’s own $P(\text{True})$, its probability of answering “True” when asked whether it was right, drops to near chance (0.47 to 0.59).

Second, test where the direction lives. Before the attempt, at the last token of the question, the model’s state is the same across all five attempts. It can encode difficulty and nothing else. A probe trained after the attempt and read at that point scores 0.49 on MMLU-Pro, which is chance, against 0.75 at its own site. The direction that separates good attempts from bad ones does not exist until the model has tried. Geometry is the exception. There the pre-attempt state already predicts failure, so that cell is mostly difficulty.

Reading before the question. There is a subtler worry. Every read so far came from a token the model reaches after being asked to assess itself. Maybe the question creates the signal rather than reveals it. So we also read the state at the last token of the answer, before any confidence question exists. On MMLU-Pro, correctness is already readable there at 0.69. From the question alone it is 0.57, so the model cannot see failure coming. Asking the model to self-assess sharpens the signal to 0.77.

Transfer across domains. Table 1 trains a probe on one domain and tests it on each of the others. On Mistral it keeps about 90% of its lift above chance (0.69 off the diagonal against 0.71 on it). The original matrix showed the same pattern in every model. The one exception is Gemma-4. There, probes trained on geometry fail elsewhere (0.31 to 0.39), while probes trained elsewhere work on geometry (about 0.70). A probe trained on the one generative task can learn features specific to it.

4 Whether the report depends on it

Present is not used. A model could carry a perfect internal record of its errors and never consult it when writing the number. So we intervene. At the confidence token, after the answer is written so the answer itself cannot change, we scale the model’s position along the correctness direction by a gain g . Gain 1 leaves the model untouched. Gain 0 erases the per-attempt information along that direction. Gains 2 and 4 exaggerate it. We run this on Mistral and MATH, and repeat it with a random direction of the same size as a control. Table 2 shows what the model wrote at each setting.

On Mistral, the report depends on it. Remove the direction ($g=0$) and stated confidence can no longer tell the model’s successes from its failures: 0.84 drops to 0.56, a change of -0.27 $[-0.42, -0.12]$. Remove the random direction and nothing happens (-0.00 $[-0.03, +0.03]$). This does not look like generic damage. The output stays well formed, and confidence on wrong answers goes *up* until it matches confidence on right answers; the model becomes uniformly sure. Turn the direction up instead and confidence on wrong answers falls while right answers hold; calibration error halves, $0.34 \rightarrow 0.19$. The same intervention

Table 2: Mistral on MATH, 150 held-out records. The correctness direction scaled by a gain at the confidence token, after the answer is written.

gain on the correctness direction	untouched 1	erased 0	doubled 2	quadrupled 4
mean confidence, wrong / right answers	84 / 96	97 / 98	70 / 94	54 / 88
AUROC of stated confidence	0.84	0.56	0.85	0.82
same, with a random direction	0.84	0.83	0.84	output broke

125 30% of the way through the network does nothing, and a direction learned on MMLU-Pro,
126 where surface features cannot explain the probe, still controls MATH’s report (0.88 $\rightarrow$ 0.71).

127 **On GLM, it does not.** GLM’s signal is the strongest in the study: within a question,
128 the probe reads 0.87 while the model’s own words read 0.53. Yet removing the direction
129 leaves stated confidence where it was, 0.76 before and after. GLM represents its errors more
130 sharply than Mistral and reports them less. That is our cleanest case of represented but not
131 reported, so the causal result is a fact about particular models, not models in general.

132 5 A second witness

133 **A label-free instrument agrees.** A trained probe invites one more objection: it might
134 have overfit to our data. So we added an instrument that uses no correctness labels. The
135 Jacobian lens [3] is fit on generic text and reads an activation for what it pushes the model
136 to say next; we score each stored state by how far it leans toward “wrong” over “correct.”

137 On dense models the lens matches the probe. On Mistral it reads 0.82 on MATH and 0.75
138 on MMLU-Pro, against the probe’s 0.83 and 0.75, with a clean input-layer control. Two
139 instruments built from different evidence find the same direction, and because the lens reads
140 through the model’s own output pathway, that direction sits where it could influence what
141 the model says.

142 **Where it fails, it shows why.** On GLM (mixture-of-experts) and Qwen3.6 (gated linear
143 attention) the lens fails (0.31 and 0.43) while the probe still works. Its “correct” and “wrong”
144 readouts collapse together, cosine 0.96 on MoE against 0.4 on dense. The lens averages one
145 map over all inputs; these architectures route differently per input, so the average smears.
146 Swap Qwen3.6 for dense Qwen2.5-14B, same family, and the lens jumps to 0.88.

147 Together: the archived matrix finds the signal in every model we tested; whether it reaches
148 the report, and whether a label-free instrument can read it, varies by model.

149 6 What this means

150 **For reliability.** Without an exact checker, the probe is the best per-attempt readout we
151 found: keeping the attempts it scores highest raises accuracy far more than keeping the ones
152 the model is most confident in, and it works where the model’s confidence is useless.

153 **Limitations.** A probe could be reading properties of failed output rather than a dedicated
154 self-assessment; the surface baseline bounds this without closing it, most loosely on MATH
155 and geometry (Appendix C) and on GLM, where surface nearly matches the probe. Every
156 number is a single seed. The before-asking test runs only on MMLU-Pro: on MATH the
157 question alone already predicts correctness at 0.81, leaving no room to see what the attempt
158 adds. The causal result holds on Mistral and fails on GLM; Gemma-4 passes 94% of MATH,
159 leaving five failures, too few to test; the Qwen3.6 run did not finish in time. Whether
160 preference tuning creates the gap is open; a base model would answer it.

161 **Takeaway.** These models carry information about their own errors that their stated
162 confidence does not report, and on one model the report depends on it.

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A The elicitation protocol, and one record end to end

This appendix gives the exact prompts behind every record, then one complete record from the Mistral-Small-24B $\times$ MATH cell, summarized first and then given in full.

One record, Mistral-Small-24B $\times$ MATH. *What integer n satisfies $0 \leq n < 18$ and $n \equiv -11213141 \pmod{18}$? The model sets the problem up correctly and gets $18 \times 622953 = 11213154$ right, then writes $r = -11213141 + 11213154 = 3$ and answers 3. That subtraction gives 13, which is the gold answer. Stated confidence: 90 before the attempt, 95 after it. Other readouts of the same attempt: P(True), the probability the model assigns to “True” when asked whether its answer is correct, 0.32; mean answer log-prob -0.05 .*

A single arithmetic slip inside a fluent derivation. Having just made the error, the model revises its stated confidence upward, while P(True) on the same transcript assigns only 0.32 probability to correctness. The rest of this appendix gives the prompts and the full transcript.

A.1 Prompts

Each record is four forward passes: the three turns of Section 2 plus one read taken beforehand, at the last prompt token of the bare question, before any elicitation. The confidence turns run with thinking off so the read lands on a clean one-line reply; the attempt turn runs with thinking on for hybrid reasoners. The fixed **Confidence**: N format is what makes the read site content-neutral: the digit token has identical local context on every record, so layer-0 activations carry no signal by construction.

System prompt (MATH).

Solve the problem step by step. Put your final answer in `\boxed{}`.

Turn 1, prospective confidence. The problem is followed by:

Before answering, estimate how likely you are to answer this question correctly. Reply with EXACTLY one line and nothing else:

205 Confidence: N
 206 where N is an integer from 0 (certain to be wrong) to 100 (certain to be
 207 correct). Do NOT answer yet.

208 **Turn 2, the attempt.** The turn-1 reply is left in context as the assistant message, then:

209 Now give your final answer inside `\boxed{}`.

210 **Turn 3, retrospective confidence.** The attempt is left in context, then:

211 Now assess the answer you just gave. On a scale of 0 to 100, how confident
 212 are you that it is correct? Reply with exactly one line and nothing else:
 213 Confidence: N

214 **P(True) baseline.** A separate teacher-forced branch off turn 2, no generation. Next-token
 215 mass on True versus False:

216 Consider your final answer above. Is it correct? Reply with exactly one word:
 217 True or False.

218 Grading is external and is never shown to the model, so the turn-3 revision is blind self-
 219 assessment.

220 **A.2 One record, verbatim**

221 Record `math_test_number_theory_753`, sample 2 of 5.

222 **Problem.** What integer n satisfies $0 \leq n < 18$ and $n \equiv -11213141 \pmod{18}$?

223 **Turn 1 reply.** Confidence: 90

224 **Turn 2 reply (complete).**

225 First, we need to find the remainder when -11213141 is divided by 18.
 226 1. Calculate $-11213141 \bmod 18$:

$$-11213141 \div 18 \approx -622952.2778$$

227 The integer part of the division is -622953 .

$$-11213141 = 18 \times (-622953) + r$$

228 where r is the remainder.
 229 2. Calculate the remainder r :

$$r = -11213141 - 18 \times (-622953)$$

$$r = -11213141 + 11213154$$

$$r = 3$$

232 Therefore,

$$-11213141 \equiv 3 \pmod{18}$$

233 So, the integer n that satisfies $0 \leq n < 18$ and $n \equiv -11213141 \pmod{18}$ is 3.

234 **Turn 3 reply.** Confidence: 95

235 **What happened.** The method is right and every intermediate is right: $18 \times 622953 =$
 236 11213154 checks out, and the model has written the correct subtraction on the line above its
 237 answer. It then evaluates $-11213141 + 11213154$ as 3 rather than 13. Gold is 13. Nothing
 238 in the presentation marks the slip. The derivation is well-formed, correctly structured, and
 239 confidently laid out, which is precisely why stated confidence rises from 90 to 95 across an
 240 attempt that failed.

241 **Readouts of this attempt.**

	readout	value
	stated confidence, prospective (turn 1)	90
	stated confidence, retrospective (turn 3)	95
242	P(True)	0.32
	mean answer log-probability	-0.05
	graded outcome	incorrect

243 This is one record, chosen to illustrate the failure mode rather than to carry evidential weight;
 244 representative cell-level numbers are in Table 3, Appendix D.

245 **B Calibration beyond ECE, and cell-level multiplicity**

246 ECE depends on binning and on sample composition, so we report two binning-free proper
 247 scoring rules alongside it. On Mistral $\times$ MATH (750 records, accuracy 0.648), the model’s
 248 retrospective stated confidence gives Brier 0.262 and NLL 1.470 against a 10-bin ECE of
 249 0.271. P(True) on the same records gives Brier 0.386: better ordered than stated confidence,
 250 worse calibrated in absolute terms. The steering results in Table 2 are reported as ECE
 251 because the steered runs stored per-gain aggregates rather than per-record confidences;
 252 recomputing Brier and NLL at each gain requires re-running the intervention and is the first
 253 thing we would add.

254 The cell-level bootstrap counts in Table 3 are uncorrected. With 16 cells, some individually
 255 significant results are expected under the null, so the cell counts should be read as descriptive.
 256 The aggregate pattern is the claim that carries weight: the effect is positive in 12 of 16 cells
 257 with one tie, significantly positive in 10, and never significantly negative. A hierarchical
 258 model pooling across cells with model and domain as random effects is the right test and is
 259 not yet run.

260 **C Alternative explanations for the probe signal**

261 A probe could key on properties of failed derivations rather than on a dedicated metacognitive
 262 variable: low token probability, hedging language, or, on MATH, an early-layer channel next
 263 to the answer text. Our surface-features baseline (answer length, digit count, line count,
 264 word count, brace count) bounds this without closing it, and the bound is domain-dependent:
 265 the probe adds +0.21 over surface on MMLU-Pro but only +0.07 on MATH and -0.11
 266 on geometry, where a construction that fails to compile is visibly malformed. Two results
 267 argue against a pure surface account on the domains where the signal is strong. A direction
 268 fitted on MMLU-Pro, where surface reaches only 0.54 against a probe of 0.75, still governs
 269 MATH’s self-report under ablation (0.880 $\rightarrow$ 0.709). And the pre-attempt read, taken where
 270 the activation is byte-identical across a question’s attempts, cannot recover the post-attempt
 271 direction on MMLU-Pro (0.490, chance), which a surface-of-the-question account would
 272 predict it could. A linear probe at one token also undercounts nonlinear or multi-token
 273 signal, so the reported numbers are a floor.

274 **D Per-cell numbers and specificity controls**

275 **E Checkpoints**

276 All models are public Hugging Face releases, run in bfloat16 without quantization,
 277 greedy decoding: mistralai/Mistral-Small-24B-Instruct-2501, Qwen/Qwen3.6-27B,
 278 zai-org/GLM-4.7-Flash, google/gemma-4-26B-A4B-it; the dense within-family control
 279 for the lens is Qwen/Qwen2.5-14B-Instruct.

Table 3: Post-attempt correctness discrimination (AUROC). Left: stated confidence vs. the probe in representative cells; overall the probe wins 12 of 16 with one tie, mean +0.09, with a paired bootstrap significantly positive in 10 of 16 and never significantly negative, uncorrected at the cell level (Appendix B). Right: Mistral specificity controls. “pre” is the last prompt token before the attempt; “post” is after the attempt; “resid.” removes the pre score; “post→pre” applies that fitted post readout at the pre-attempt site, where only difficulty is visible.

model × domain	stated	probe	Mistral	pre	post	resid.	post→pre
Gemma-4 × MATH	0.57	0.78	MMLU-Pro	0.568	0.754	0.751	0.490
Qwen3.6 × MMLU-Pro	0.67	0.84	MATH	0.792	0.834	0.701	0.624
GLM × MMLU-Pro	0.67	0.85	GPQA	0.593	0.572	0.544	0.479
Mistral × MATH	0.79	0.83	GeoGen	0.770	0.676	0.505	0.648